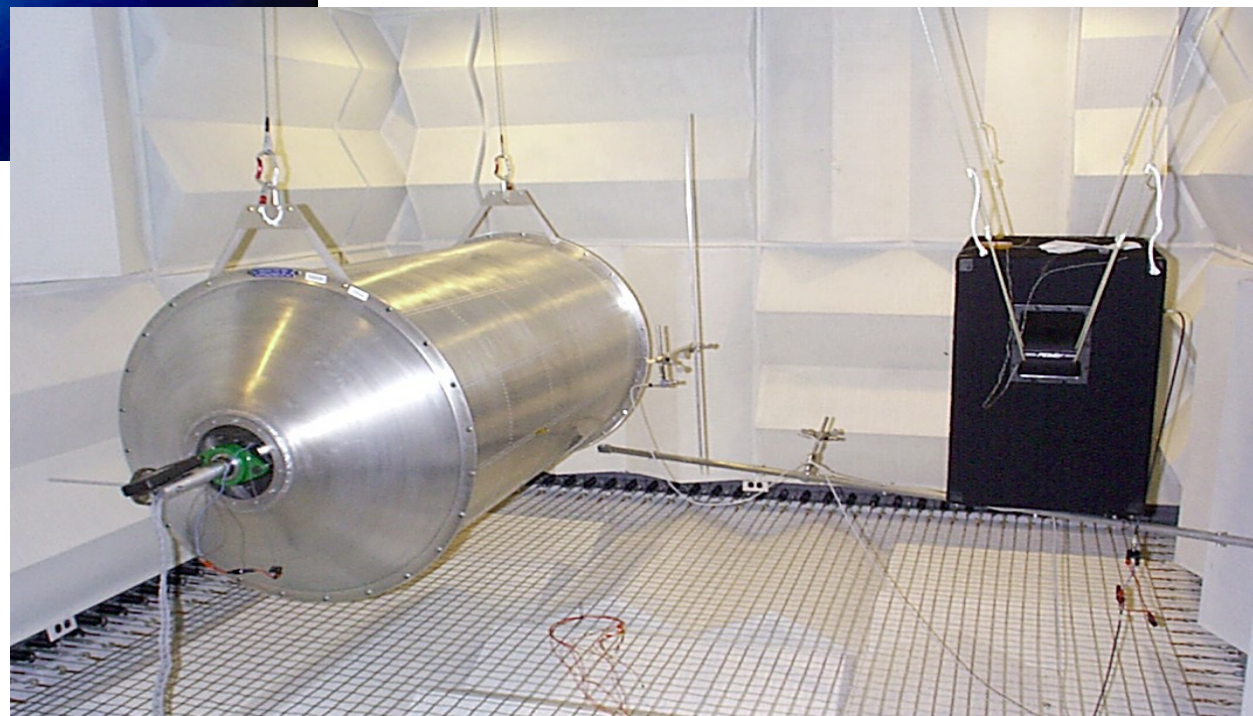
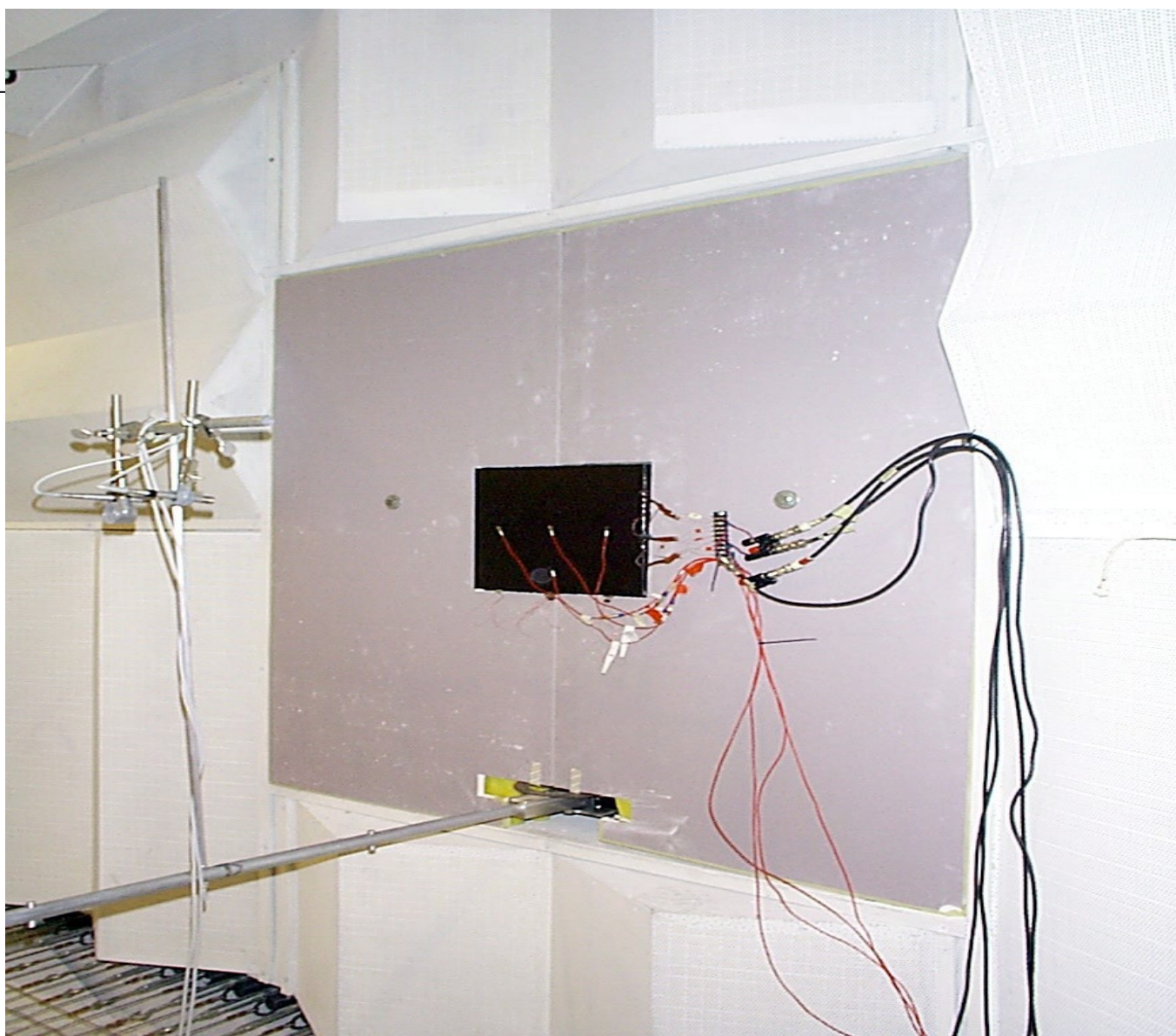


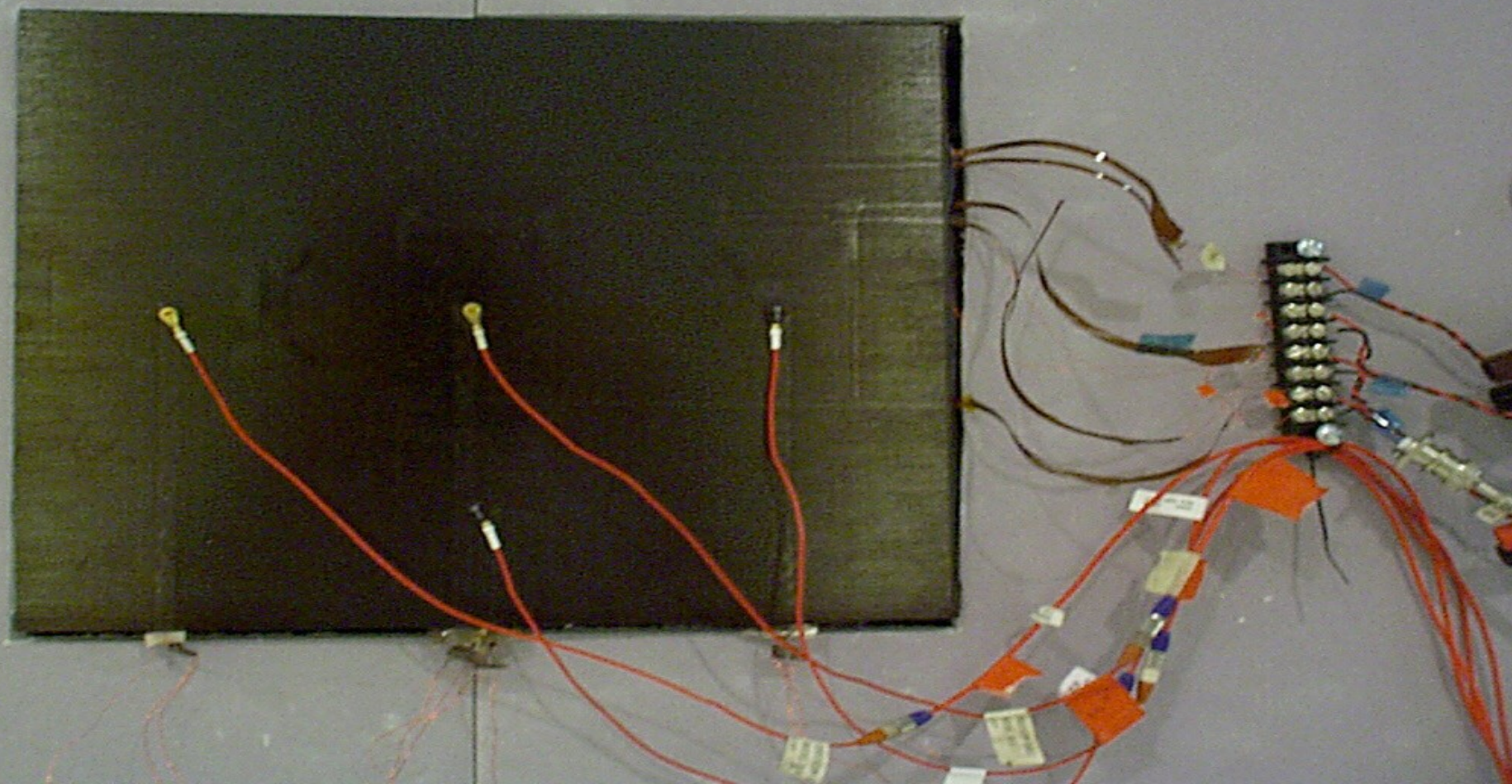
ME 2801

Introduction to Control Systems

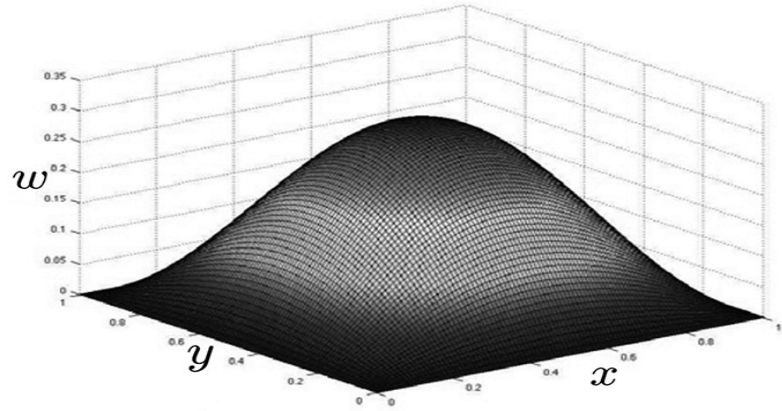
W8: Frequency Response,
Bode Plot Introduction



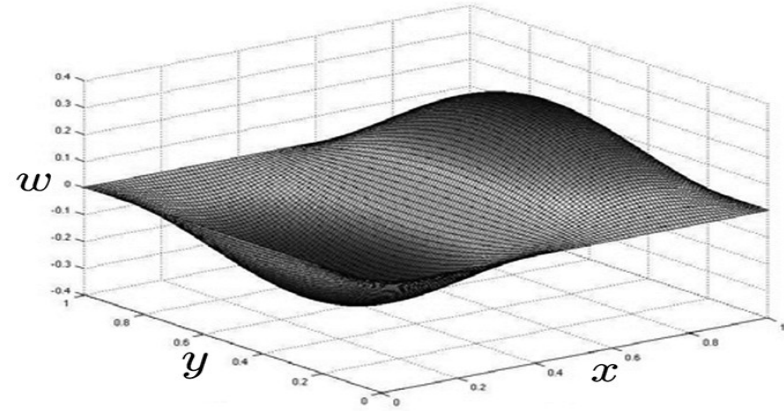




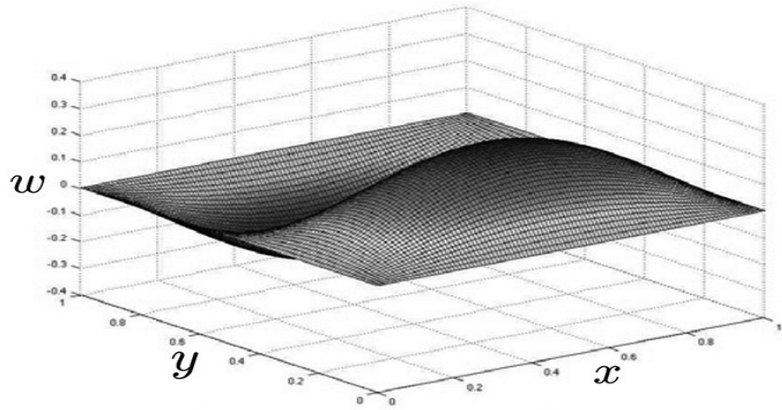
Modes of Vibration



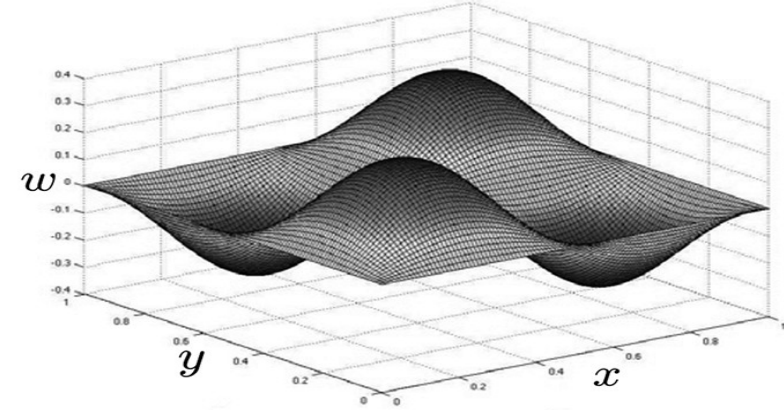
(a) Mode 1



(b) Mode 2

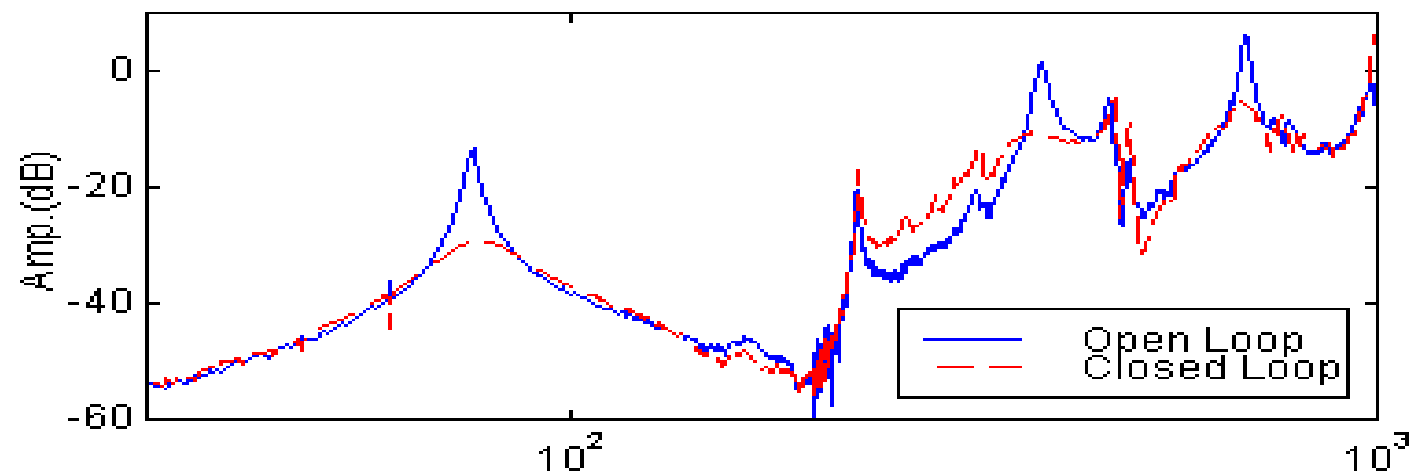


(c) Mode 3

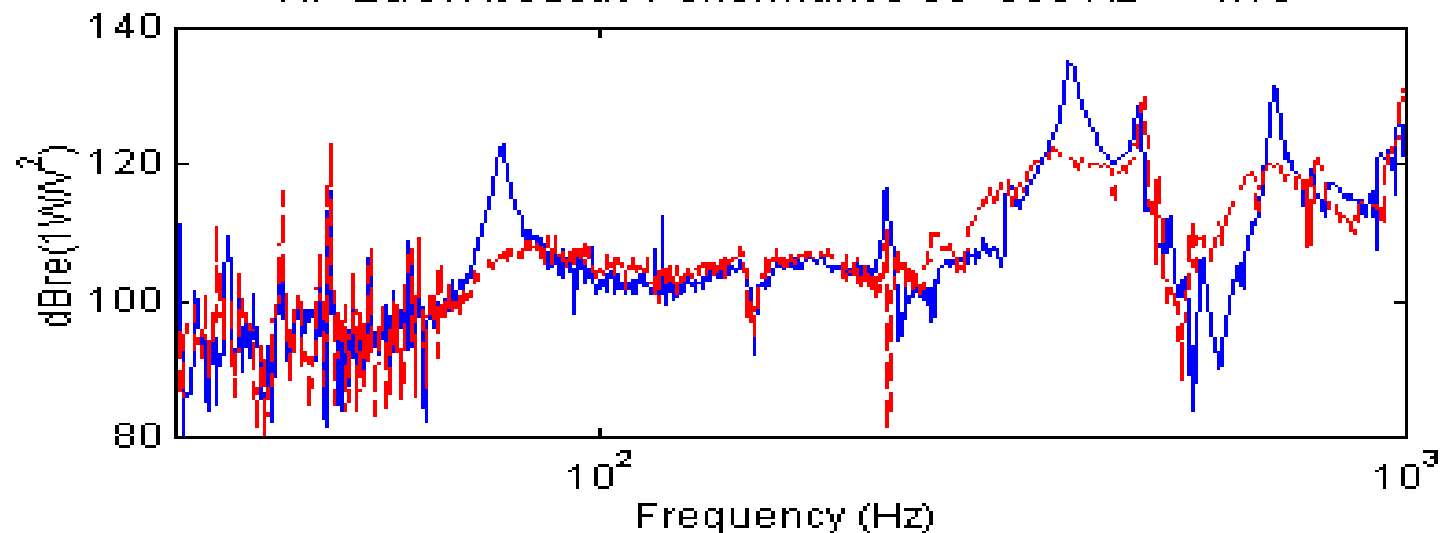


(d) Mode 4

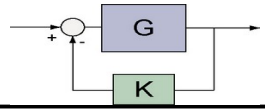
RF-LQG: Vibration Performance 30- 900 Hz = 4.27



RF-LQG: Acoustic Performance 30- 900 Hz = 4.13

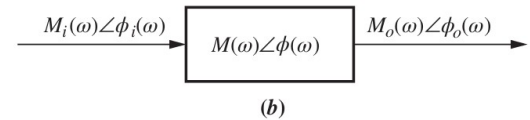
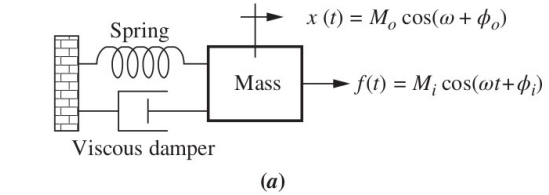


Frequency Response for ME2801



Concept of frequency response

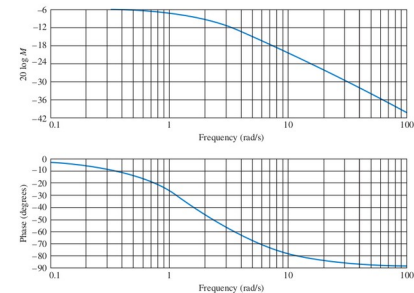
- Steady-state output for sinusoidal input
- Gain and phase response



Analysis

- From transfer function to frequency response function
- Bode Plot display
 - Sketching using straight-line approximation
 - Building blocks approach
- Performance and stability (margins) in frequency domain

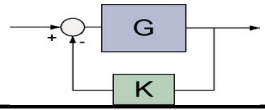
$$G(s) = 1/(s + 2) \rightarrow$$



Design

- Gain compensation to meet performance and stability goals

Frequency Response



Another design tool for predicting and crafting the system performance.

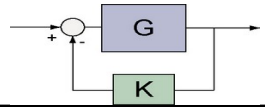
Root Locus

- Design in the s-domain is about crafting the **transient response**
- About location of (dominant) closed-loop poles
- Requires transfer function model – poles and zeros
- Can't “see”
 - Stability margins

Freq. Resp.

- Design with Bode plots is about crafting the **steady-state response**
- About gain and phase of the output as a function of input frequency.
- Can use empirical / uncertain plant models
- Can “see” some new things
 - sensitivity to noise, margins of stability, bandwidth

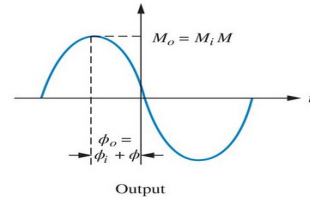
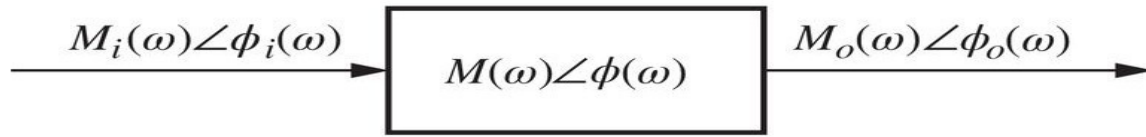
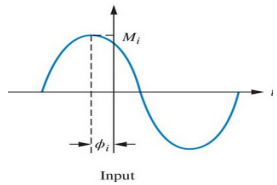
Control design tools are complementary –
often need to use both!



For an LTI system – put in a sinusoid, get out sinusoid

$$x(t) = X \sin(\omega t) \quad \frac{x(t)}{X(s)} \rightarrow \boxed{G(s)} \rightarrow \frac{y(t)}{Y(s)} \quad y(t) = |G(j\omega)| X \sin(\omega t + \varphi)$$

- **Same Frequency!**



A transfer function consists of gain and phase

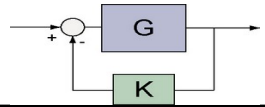
- Gain/Magnitude: amplification / attenuation

$$M(\omega) = \frac{M_o(\omega)}{M_i(\omega)} \rightarrow M_o(\omega) = M(\omega) M_i(\omega)$$

- Phase: time-shift

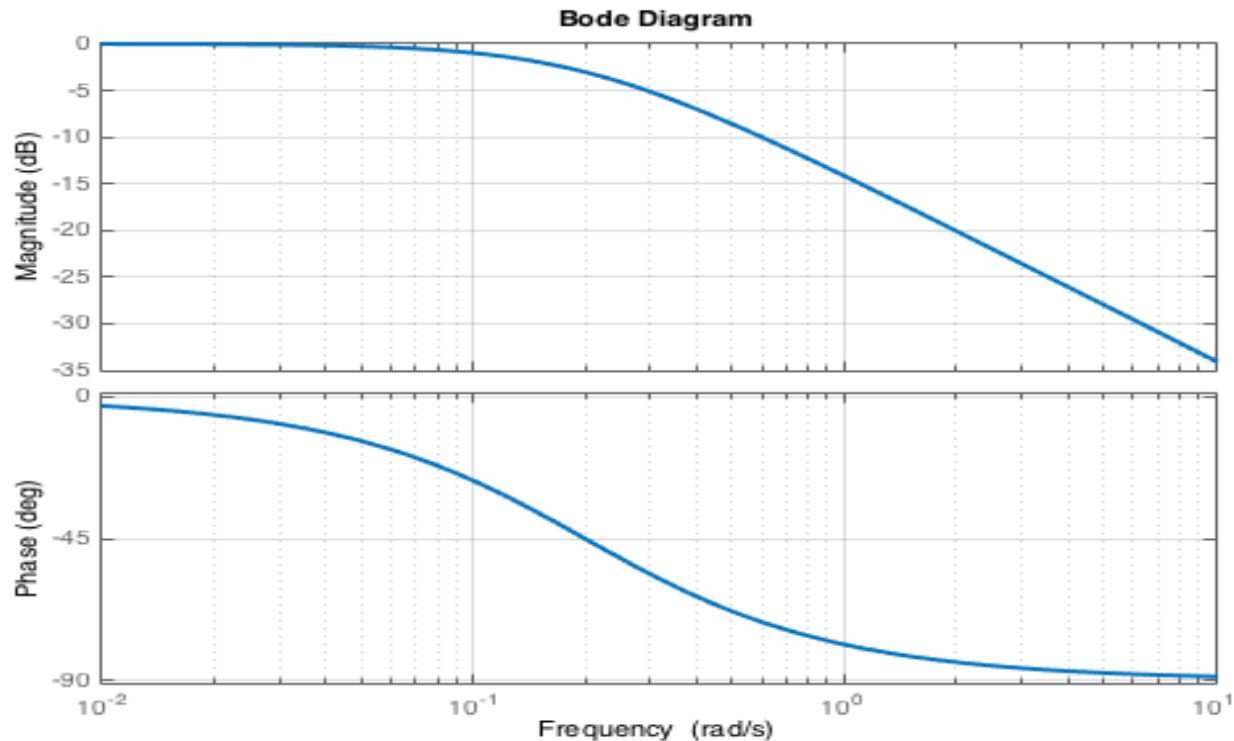
$$\varphi(\omega) = \varphi_o(\omega) - \varphi_i(\omega) \rightarrow \varphi_o(\omega) = \varphi_i(\omega) + \varphi(\omega)$$

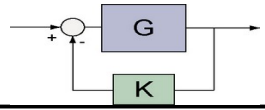
Bode Plot Overview



Bode = 2 plots

- Magnitude vs. Freq on semilogx scale
 - Magnitude is typically logarithmic as well
- Phase vs Freq on semilogx scale
- Relatively simple to sketch
- Multiplication becomes addition thanks to logarithms!

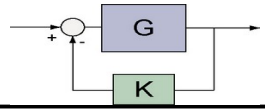




Definition

$$|G| \text{ [dB]} = 20 \log_{10}(|G|) = 20 \log_{10}\left(\frac{\text{out}}{\text{in}}\right)$$

Ratio $ G $	$ G \text{ [dB]}$
1	
10	
100	
0.1	
2	
0.5	
20	
0.05	



Definition:

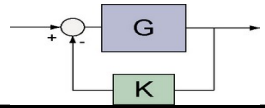
$$|G| \text{ [dB]} = 20 \log_{10}(|G|) = 20 \log_{10}\left(\frac{\text{out}}{\text{in}}\right)$$

Recall:

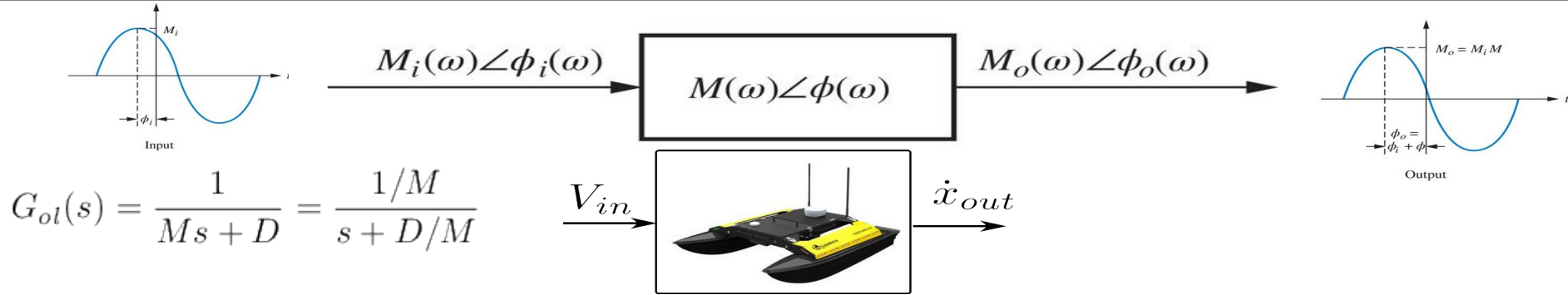
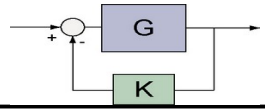
$$\log(A B) = \log(A) + \log(B) \text{ and } \log(A^n) = \log(n) \log(A)$$

Ratio $ G $	$ G \text{ [dB]}$
1	0
10	20
100	40
0.1	-20
2	6
0.5	-6
20	26
0.05	-26

Frequency Response: Magnitude

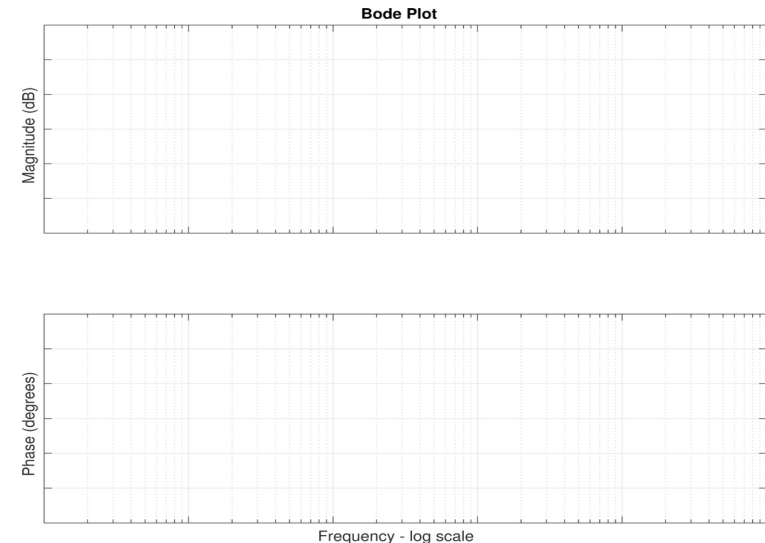


Example

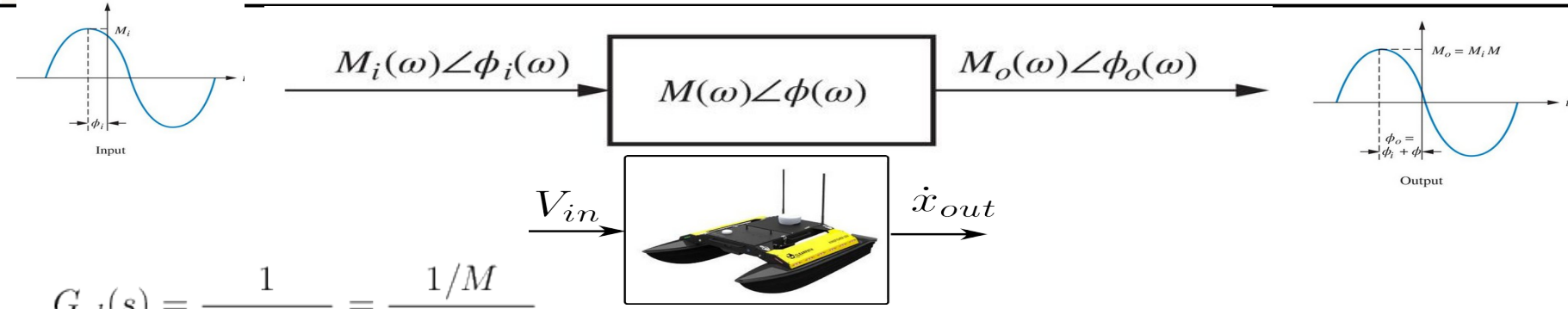
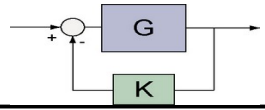


Consider an experiment where we...

- Input a sinusoidal voltage, varying the thrust
- Measure the output sinusoidal velocity
- Plot the frequency response points on a Bode paper

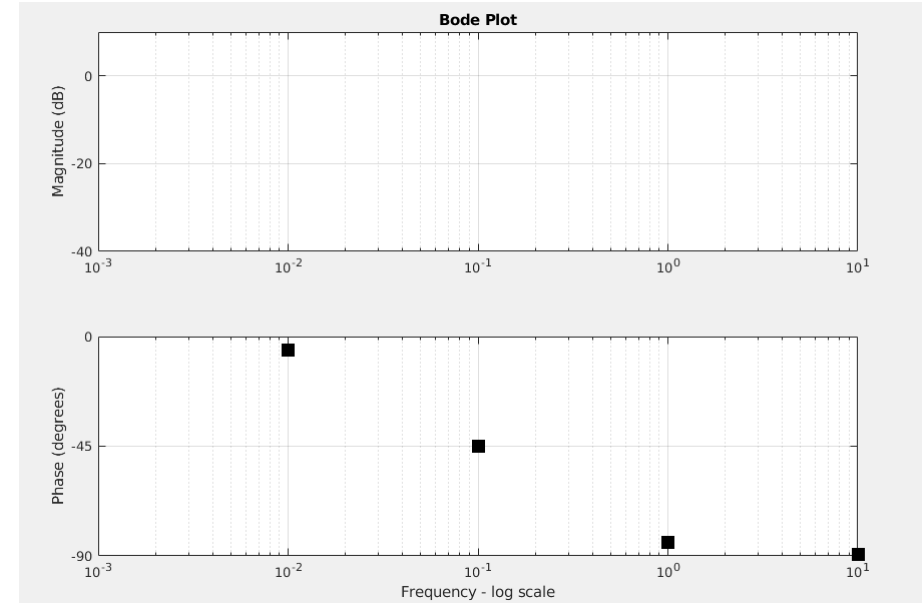


USV Experimental Data

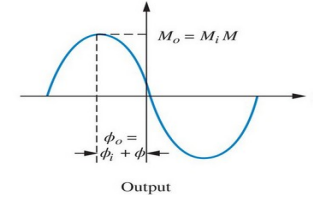
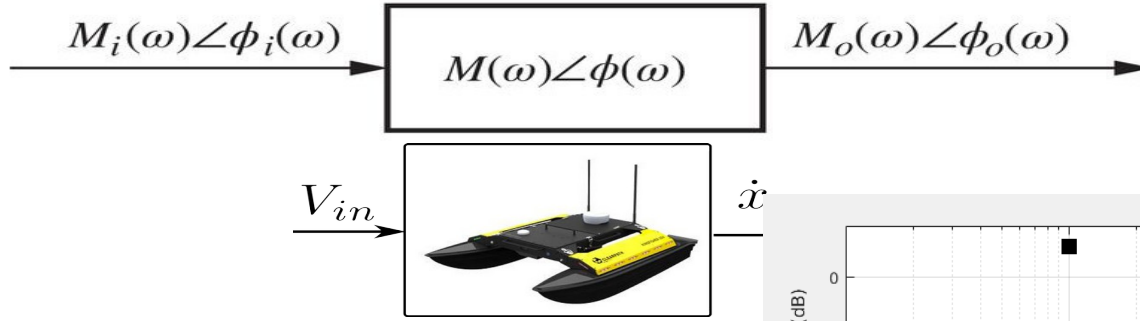
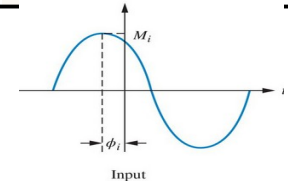
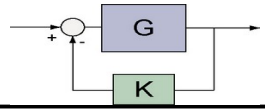


$$G_{ol}(s) = \frac{1}{Ms + D} = \frac{1/M}{s + D/M}$$

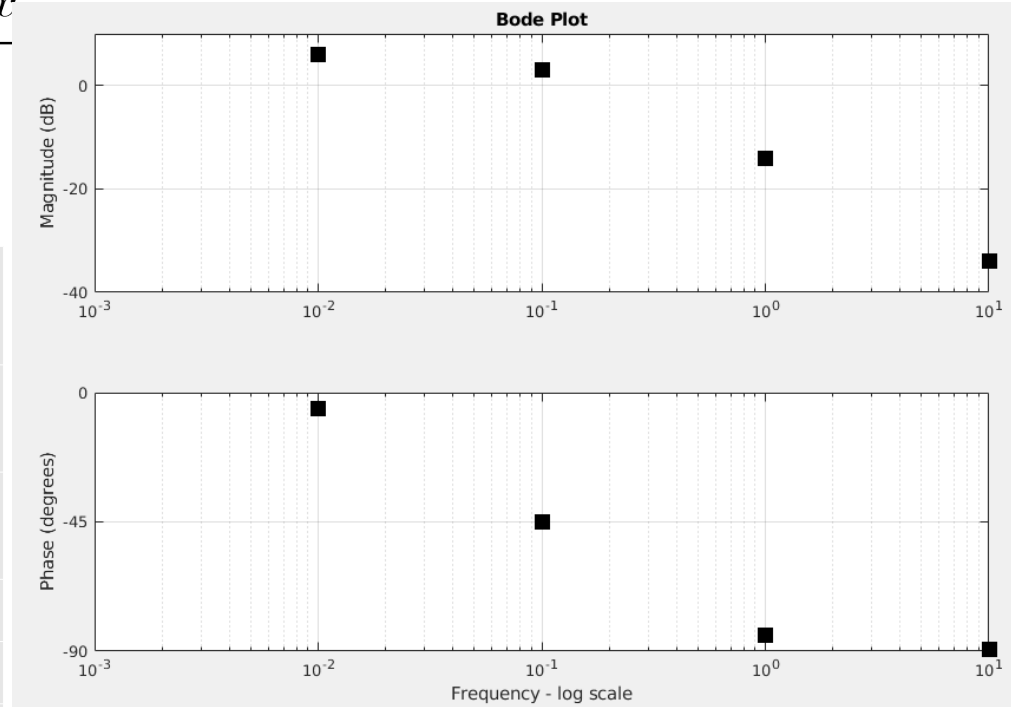
Frequency [rad/s]	Input Voltage		Output Speed		
	Amplitude [V]	Phase [deg]	Amplitude [m/s]	Phase [deg]	
0.01	1.0	0	2.0	0	
0.1	1.0	0	1.0	-45	
1.0	1.0	0	0.2	-90	
10.0	1.0	0	0.02	-90	



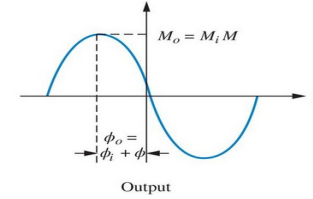
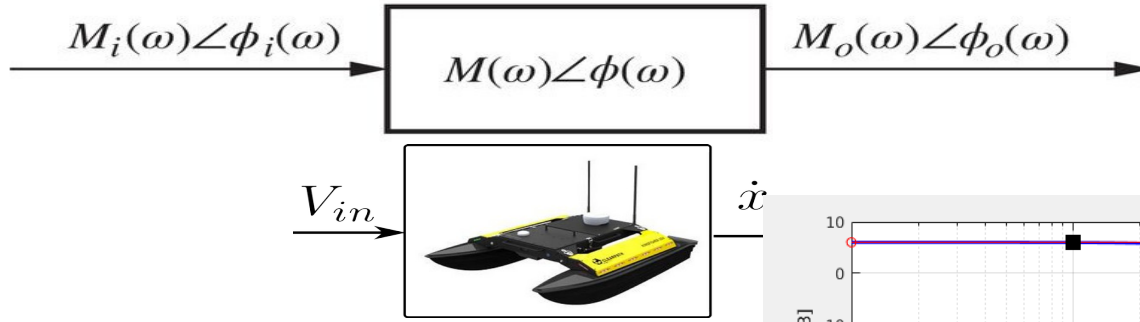
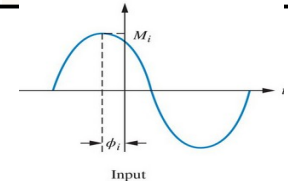
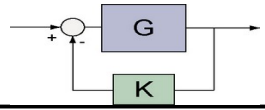
USV Experimental Data



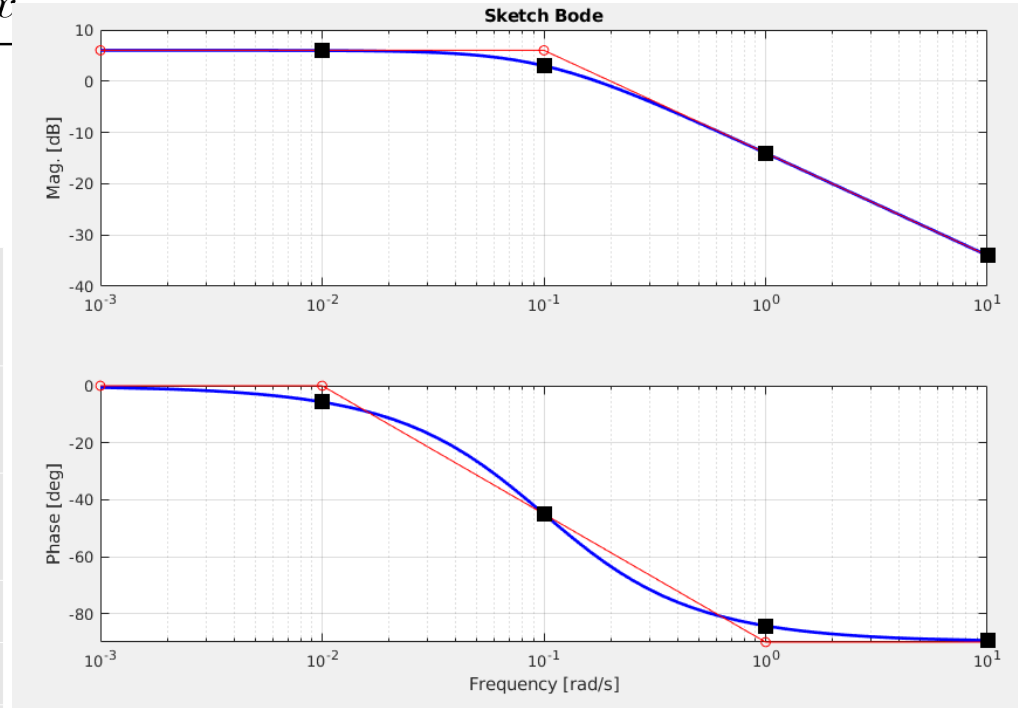
Frequency [rad/s]	Input Voltage		Output Speed		Magnitude Frequency Response
	Amplitude [V]	Phase [deg]	Amplitude [m/s]	Phase [deg]	Ratio / dB
0.01	1.0	0	2.0	0	2.0 / 6 dB
0.1	1.0	0	1.0	-45	1.0 / 0 dB
1.0	1.0	0	0.2	-90	0.2 / -14 dB
10.0	1.0	0	0.02	-90	0.02 / -34 dB

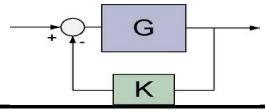


USV Experimental Data



Frequency [rad/s]	Input Voltage		Output Speed		Magnitude Frequency Response
	Amplitude [V]	Phase [deg]	Amplitude [m/s]	Phase [deg]	Ratio / dB
0.01	1.0	0	2.0	0	2.0 / 6 dB
0.1	1.0	0	1.0	-45	1.0 / 0 dB
1.0	1.0	0	0.2	-90	0.2 / -14 dB
10.0	1.0	0	0.02	-90	0.02 / -34 dB





- Bode Building Blocks
 - S-terms, first-order and second-order
- Process for sketching diagrams from building blocks
- Examples